

# Subcontracting Desk

From a group-chat decision  
to a checked, recorded allocation

## Group NN

屈妍玥 Qu Yanyue · 张锦若 Zhang Jinruo · 吴杰 Wu Jie  
李岩 Li Yan · 吴若晗 Wu Ruohan · 许佩瑶 Xu Peiyao

[github.com/dss5105-track2/track2-desk](https://github.com/dss5105-track2/track2-desk) · 16 October 2026

### THE CHAT MESSAGE

*“Send it to QuickStitch, need it back by the 19th.”*



### THE DESK'S ANSWER

## GiantWeave

back Fri 10 Apr, on time · Sat 11 Apr if reworked

QuickStitch not considered: excluded by Boss  
Every number traced to a tool output

# A batch goes outside on one chat message: unchecked and unrecorded

## TODAY

*"@Chen — order 042, 1,500 crewnecks. Send it to QuickStitch, need it back by the 19th."*

- 1 **Can they make it?** Not every workshop makes every product.
- 2 **Can they take it now?** A fast shop with a 3-day queue is not fast today.
- 3 **Are they allowed to?** One shop is suspended after an audit, one is on trial with small batches only.

*Get one wrong and the batch bounces back a day later. In a ten-day cycle that is the margin gone.*

## WITH THE DESK

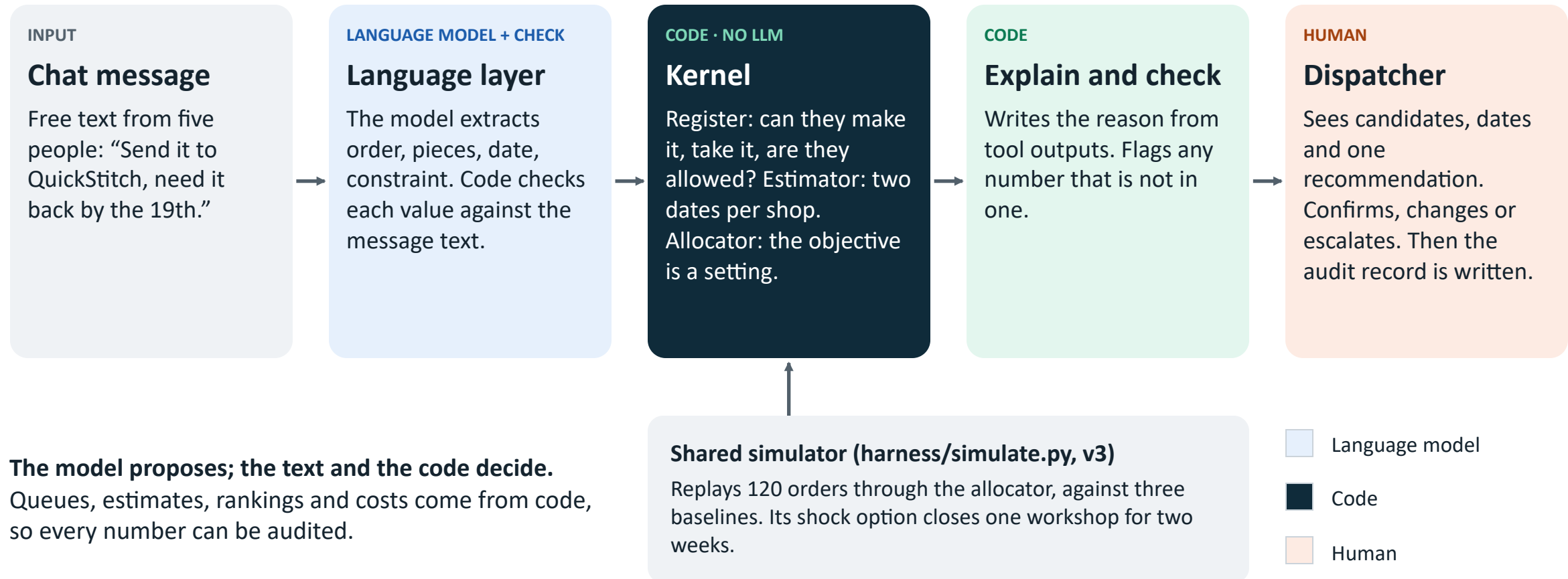
R12 · ORD-053 · 800 polo shirts, due Fri 10 Apr

Standing constraint recorded: Boss, nothing new to QuickStitch this week

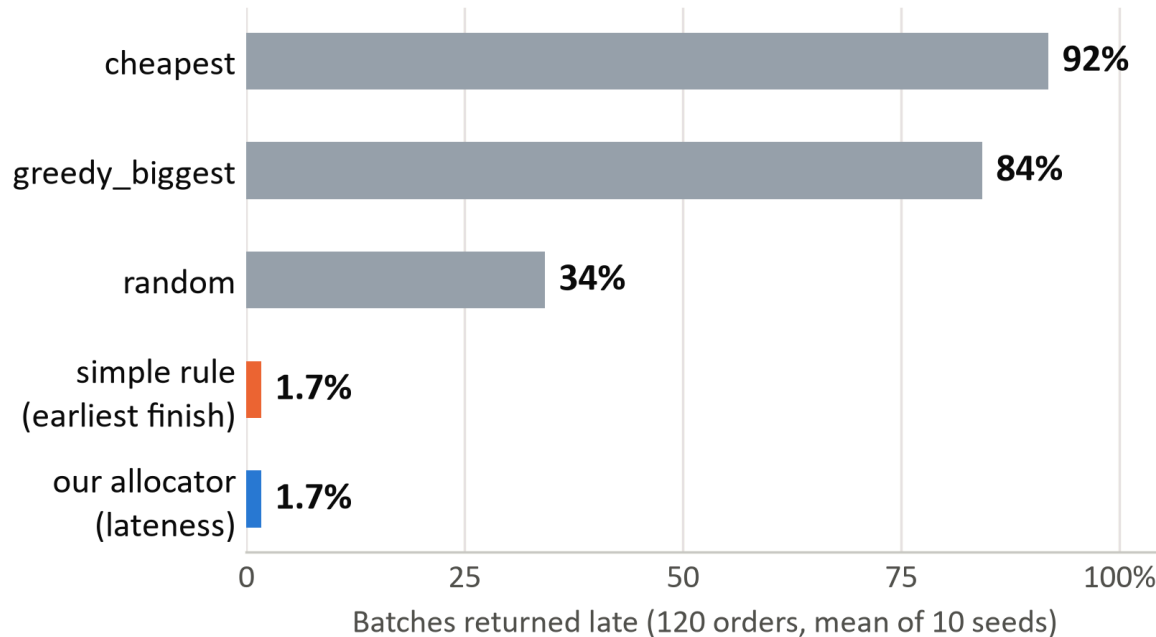
|   | Workshop      | Back       | If reworked | Status   |
|---|---------------|------------|-------------|----------|
| ★ | GiantWeave    | Fri 10 Apr | Sat 11 Apr  | on time  |
|   | Nimble Needle | Thu 9 Apr  | Sun 12 Apr  | on time  |
|   | SteadyHands   | Fri 10 Apr | Mon 13 Apr  | on time  |
|   | BudgetWorks   | Mon 13 Apr | Tue 14 Apr  | 3 d late |

**Why GiantWeave:** it only makes tops, so the two shops that also make accessories stay free for tighter batches.  
Not eligible: QuickStitch (excluded), OldMill (suspended), FreshStart (300-piece cap), Little Loom (accessories only).

# The language model reads and explains; the arithmetic stays in code



# In replay, late batches fall from 34% to 1.7%



Source: harness/simulate.py v3, seeds 5105 to 5114, repo commit 1c6cb61. Every order is replayed as if it had gone outside on its order date.

## WHAT THE DISPATCHER GETS

**9.3 days**

mean turnaround, down from 21.9 days when workshops are picked at random

**+3.3% cost**

against random (115.4k against 111.8k): speed is not free

**17 of 30**

morning messages cannot simply be dispatched: 7 need a question, 4 name a workshop that cannot take the job, 3 cannot meet the date, 3 have no answer in the data

# Sprint 1 so far: the loop runs end to end, and each part is now measured

To confirm (13)

Needs a decision (7)

Ask requester (7)

Can't answer (3)

Done (0)

Confirm 7 safe recommendations in one click

Held back for you: 3 tight (a rework would make them late), listed first, and 3 that would take a workshop a tight one needs.

|                          | ID  | Order   | Decision             | Back       | Spare |
|--------------------------|-----|---------|----------------------|------------|-------|
| <input type="checkbox"/> | R12 | ORD-053 | ★ GiantWeave · tight | Fri 10 Apr | 0 d   |
| <input type="checkbox"/> | R17 | ORD-017 | ★ GiantWeave · tight | Fri 10 Apr | 0 d   |
| <input type="checkbox"/> | R21 | ORD-081 | ★ GiantWeave · tight | Sun 12 Apr | 1 d   |
| <input type="checkbox"/> | R25 | ORD-109 | ★ GiantWeave         | Thu 09 Apr | 4 d   |
| <input type="checkbox"/> | R09 | ORD-045 | ★ Nimble Needle      | Sat 04 Apr | 5 d   |
| <input type="checkbox"/> | R06 | ORD-096 | ★ Nimble Needle      | Sat 11 Apr | 6 d   |
| <input type="checkbox"/> | R16 | ORD-024 | ★ FreshStart         | Mon 06 Apr | 9 d   |
| <input type="checkbox"/> | R04 | ORD-041 | ★ Little Loom        | Sat 11 Apr | 10 d  |
| <input type="checkbox"/> | R22 | ORD-040 | ★ GiantWeave         | Sat 11 Apr | 10 d  |
| <input type="checkbox"/> | R23 | ORD-005 | ★ Little Loom        | Fri 10 Apr | 10 d  |
| <input type="checkbox"/> | R26 | ORD-010 | ★ GiantWeave         | Sat 11 Apr | 14 d  |
| <input type="checkbox"/> | R27 | ORD-095 | ★ SteadyHands        | Fri 10 Apr | 16 d  |
| <input type="checkbox"/> | R07 | ORD-063 | ★ FreshStart         | Sat 04 Apr | 25 d  |

New message

Boss

e.g. ORD-066 — 1000 polo shirts, due Apr 20

Send to desk

R12 · 08:48 · Ravi

Confirm

ORD-053 — 800 polo shirts for UrbanThread, due Apr 10. Boss says nothing new to QuickStitch this week, they're swamped.

ORD-053 - 800 pcs TOPS - due Fri 10 Apr (9 days) - avoid QuickStitch - parsed by rules

★ GiantWeave · back Fri 10 Apr · on time, 0 days to spare · if reworked: Sat 11 Apr (late) · cost 880

Why: on time, though a rework would make it late (no eligible shop is safe against that); it only makes TOPS, so the shops that can take other work stay free for tighter batches.

☒ every number comes from a tool output · 4 tool calls

• session constraint recorded: exclude QuickStitch (by Boss)

Eligible workshops · ranked by lateness

|   | Workshop      | Back       | Days to spare | If reworked       | Days | Queue | Wo |
|---|---------------|------------|---------------|-------------------|------|-------|----|
| ★ | GiantWeave    | Fri 10 Apr | 0             | Sat 11 Apr (late) | 9.1  | 3     |    |
|   | Nimble Needle | Thu 09 Apr | 1             | Sun 12 Apr (late) | 8.3  | 1     | €  |
|   | SteadyHands   | Fri 10 Apr | 0             | Mon 13 Apr (late) | 9.3  | 2.5   | 4  |
|   | BudgetWorks   | Mon 13 Apr | -3            | Tue 14 Apr (late) | 11.6 | 5     | 3  |

Not eligible: QuickStitch (excluded by request) · Little Loom (cannot make TOPS (equipped for ACCESSORIES)) · OldMill (status=SUSPENDED) · FreshStart (batch of 800 exceeds its 300-piece limit)

Confirm GiantWeave

Nimble Needle · back Thu 09 Apr

Send to Nimble Needle

Dispatcher interface: grouped by next step, tightest first; one click confirms the rework-safe recommendations

DSS5105 Capstone · Track 2 · Subcontracting Desk · Sprint 1

56 / 56

Tests pass. Kernel, dispatcher workflow, the new rule, and the checks on what the model extracts.

30 / 30

Replies fully traced. Every number in every reply on the 30-message inbox comes from a tool output.

κ 0.70

Six members labelled the inbox on their own. The 9 changed requests: unanimous. Weakest where the named workshop cannot take the job.

86% → 100%

Right behaviour with GPT-5 nano on the 30 we developed on. A sharper prompt, plus code that checks each extracted value against the message. 97% on 38 new messages.

5

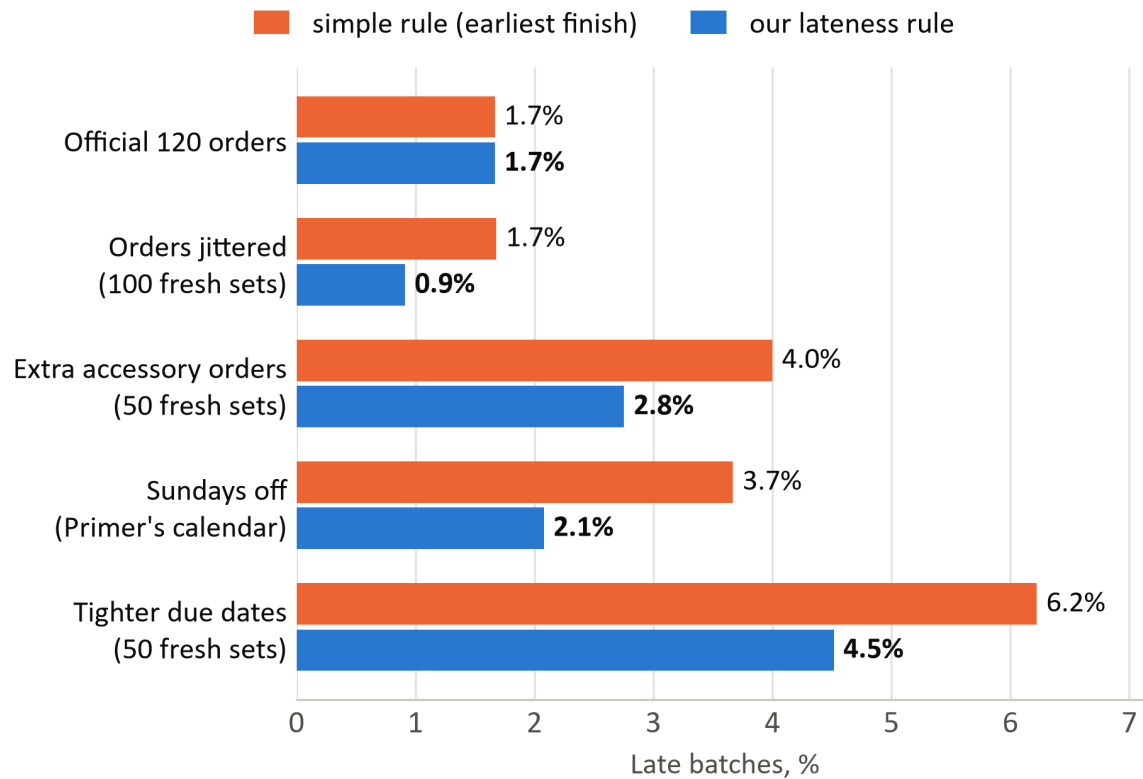
# Our lateness rule ties the simple rule on the official data, and wins when the data moves

## PRIMARY OBJECTIVE

## Minimise lateness

**The rule, in order:** on time even if the batch is reworked; then the least flexible shop (a specialist before one that makes both categories); then earliest finish.

- **Official data:** a tie on late batches (1.7%), half the late days (0.09 against 0.18).
- **Fresh data:** better on 72 of 100 jittered order books, worse on 3.
- **Where the simple rule still wins:** mean turnaround, 8.3 against 9.3 days.



Official: seeds 5105 to 5114. Fresh sets were generated after the rule was fixed (seeds 20000+), 3 seeds each. The allocators are not told about Sundays.

# What could go wrong, and what we are assuming

## RISKS

**Our labels are ours, not the instructor's.** Frozen on 7 Oct: 5 of 30 fell short of 5/6 agreement and were settled by a team vote on one stated rule. The counts per kind match the instructor's.

**On the official data we only tie the simple rule.** Our edge shows on jittered and stressed data. We report both, and where the baseline wins.

**The cheap model misreads messages.** GPT-5 nano: 86% right behaviour before our checks. A teammate-written held-out inbox is still to come.

**The simulator is a what-if replay.** With Sundays off, the simple rule goes from 1.7% to 3.7% late. We report that separately.

## ASSUMPTIONS

**Today is 1 April 2026; requests arrive in time order.** A confirmed dispatch occupies capacity for later requests (team decision).

**Standing constraints persist.** “Nothing new to QuickStitch this week” applies to every later request, with who set it.

**Two dates per estimate.** Expected: queue + work + expected rework + transport. If reworked: work takes half as long again (simulator rule).

**The data has no customer prices and no history.** Questions about either are declined, never invented.

# Sprint 2, 30 October: from a working loop to evidence we can defend

5 – 9 Oct

## Align and freeze

Gold labels frozen on 7 Oct;  
round-2 code to merge.

10 – 16 Oct

## Sprint 1

Rehearse twice; present live on  
16 Oct.

17 – 23 Oct

## Evidence

Held-out inbox written and  
frozen; user test with 3 people;  
closed-workshop detection.

24 – 29 Oct

## Ship

Public demo; Evaluation draft  
with 10 failure cases; 5-minute  
video.

### SUCCESS CRITERIA · MEASURED ON v3 DATA

**$\leq 2\%$**

late batches on the official  
orders, normal and shock (10  
seeds). Now 1.7% and 2.0%.

**Beat it**

fewer late batches than the  
simple rule on fresh order  
books. Now 0.9% against 1.7%.

**$\geq 90\%$**

right behaviour on a held-out  
inbox written by teammates,  
labels frozen first. Now 97%  
(nano) and 100% (mini) on 38  
AI-written messages.

**100%**

of numbers traced; promised  
rework dates kept (now 100%,  
99% under shock).

# Appendix A - Results on v3 data, mean of 10 seeds

| Policy                                        | Late %     | Late days   | Mean days  | Defective pcs % | Max share % | Cost           | Late % in shock |
|-----------------------------------------------|------------|-------------|------------|-----------------|-------------|----------------|-----------------|
| random (official)                             | 34.2       | 6.13        | 21.9       | 5.6             | 23          | 111,780        | 38.2            |
| greedy_biggest (official)                     | 84.2       | 26.53       | 50.3       | 5.3             | 70          | 108,430        | 84.9            |
| cheapest (official)                           | 91.8       | 81.92       | 106.2      | 8.2             | 70          | 85,960         | 91.8            |
| <b>earliest_finish (simple rule)</b>          | <b>1.7</b> | <b>0.18</b> | <b>8.3</b> | <b>7.2</b>      | <b>26</b>   | <b>115,566</b> | <b>1.9</b>      |
| <b>lateness: specialist-first rule (ours)</b> | <b>1.7</b> | <b>0.09</b> | <b>9.3</b> | <b>7.7</b>      | <b>28</b>   | <b>115,438</b> | <b>2.0</b>      |
| use_slack: latest safe finish (ours)          | 1.0        | 0.07        | 14.0       | 6.8             | 32          | 105,502        | 1.7             |
| lateness, round 1 (ours)                      | 1.7        | 0.18        | 8.2        | 7.5             | 26          | 115,592        | 1.9             |
| hybrid (ours)                                 | 4.3        | 0.29        | 10.1       | 6.1             | 29          | 111,238        | 5.5             |
| defects (ours)                                | 28.7       | 2.23        | 21.1       | 5.1             | 43          | 110,396        | 31.0            |
| fairness (ours)                               | 6.8        | 0.64        | 12.6       | 7.1             | 20          | 113,088        | 10.1            |
| cost (ours)                                   | 4.4        | 0.24        | 17.2       | 5.7             | 38          | 100,931        | 5.2             |

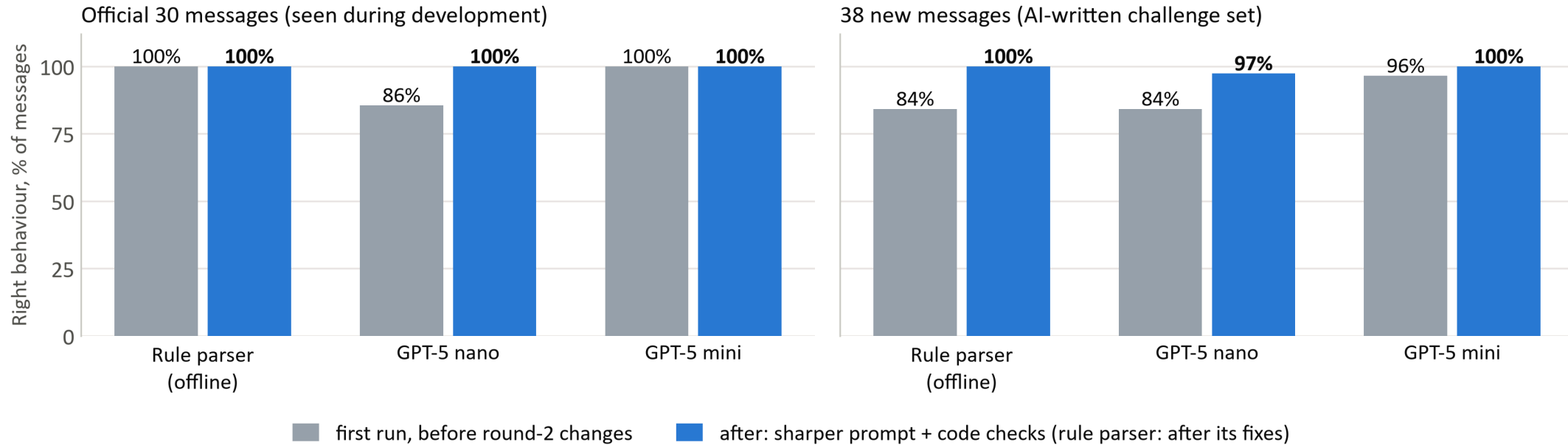
simulate.py v3, commit 1c6cb61. use\_slack is late less often and costs less, but takes 14 days on average. Single-seed defect draws move defective pcs by about  $\pm 2$  points.

# Appendix B · How the desk sorted the 30 messages



★ changed by the instructor in v3. Shown with the morning queues; the counts by kind match the instructor's published mix. Processed in order, 11 of the 13 on-time requests stay on time (R21 is 1 day late; R25 is 1 day late); at most 12 can.

# Appendix C · The language layer, measured on two inboxes



- **The rule parser was built around the 30:** 100% there, 84% on new wording at first try.
- **Model choice matters:** with the round-1 prompt GPT-5 nano gets 86% of the 30 right, GPT-5 mini 100%. After our checks nano still misreads one new phrasing (a date “sooner than planned, not said when”) in 3 runs of 3.
- **Not a clean test yet:** the 38 new messages are AI-written and two of our fixes came from errors seen on them. The held-out inbox will be written by teammates.

# Appendix D · Promised dates, and a six-day week

QUOTING TWO DATES · 1,200 BATCHES, LATENESS RULE

94%

of expected dates hold. Every miss in normal runs is a defect rework.

100%

of “if reworked” dates hold (99.2% with a workshop closed). They are 1.8 days later on average.

SUNDAYS OFF, AS THE PRIMER SAYS · LATE BATCHES %

| Policy                                | Seven-day week | Sundays off |
|---------------------------------------|----------------|-------------|
| random                                | 34.2           | 50.2        |
| earliest_finish (simple rule)         | 1.7            | 3.7         |
| lateness rule (ours)                  | 1.7            | 2.1         |
| lateness rule, estimate knows Sundays | 1.7            | 1.8         |

Same orders, same defect draws; only change: no work on Sundays. The allocators are not told. With the Sunday rule switched off the variant reproduces simulate.py exactly.